UNIVERSITY OF GUJRAT

Introduction to Photogrammetry

Lecture 08

Camera Calibration and Interior Orientation

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In Today’s Class

1. Introduction

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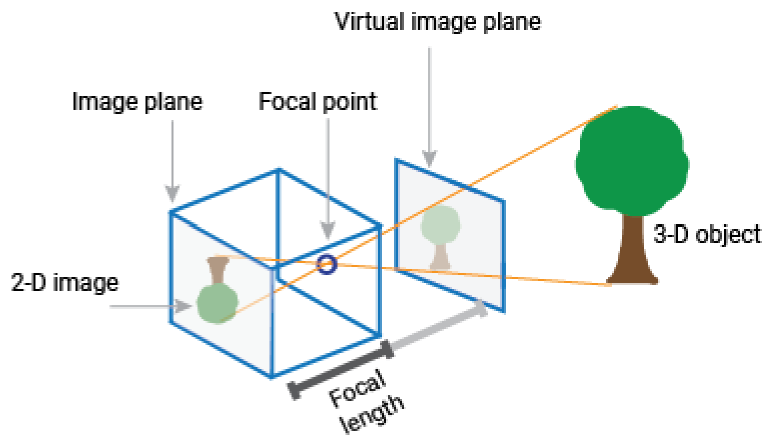
- Photogrammetry depends on accurate measurements from photographs.
- To make these measurements reliable, the internal geometry of the camera must be known.
- Camera calibration determines the geometric properties of the camera system.
- These parameters are later used in photogrammetric calculations.

What is Camera Calibration?

- Camera calibration is the process of determining the **internal characteristics of a camera**.
- It identifies the exact geometric relationship between the camera lens and the image plane.
- Calibration also measures lens distortions and other systematic errors.
- The results of calibration are used to correct image measurements.



2. Interior Orientation

- Interior orientation refers to the **internal geometry of the camera**.
- It describes the relationship between the camera lens and the photograph.
- Interior orientation parameters help recreate the geometry of the camera during image processing.
- This step is essential in photogrammetric analysis.

Main Interior Orientation Parameters

- The most important interior orientation parameters include:
  - Principal point location
  - Focal length
  - Coordinates of fiducial marks
  - Lens distortion parameters
- These parameters define the geometry of the camera system.

Principal Point

- The **principal point** is the point where the camera's optical axis intersects the image plane.
- It represents the center of the photograph from a geometric perspective.
- The position of the principal point is determined during camera calibration.
- Accurate knowledge of this point is essential for photogrammetric calculations.

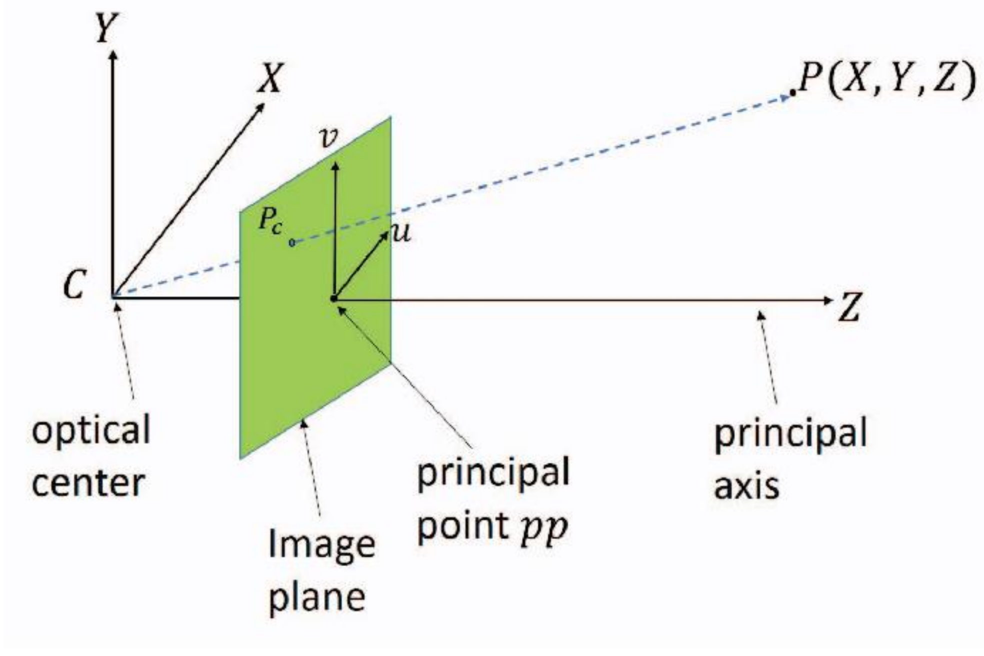
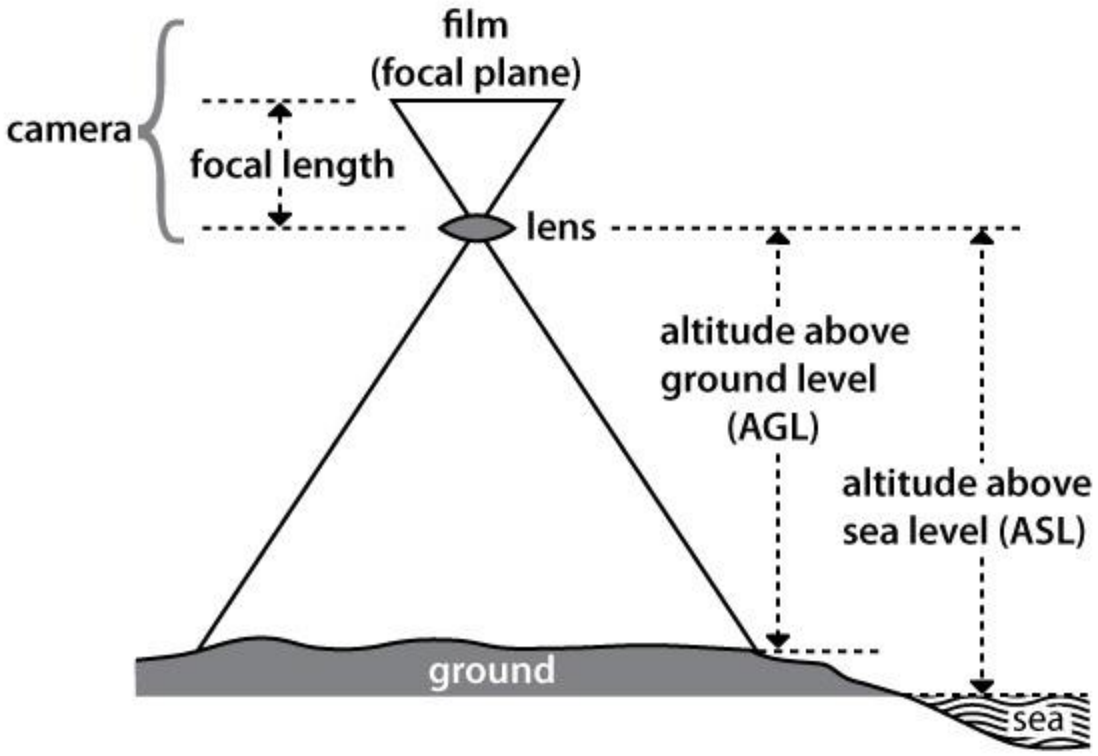


Fig. 1: Camera calibration model

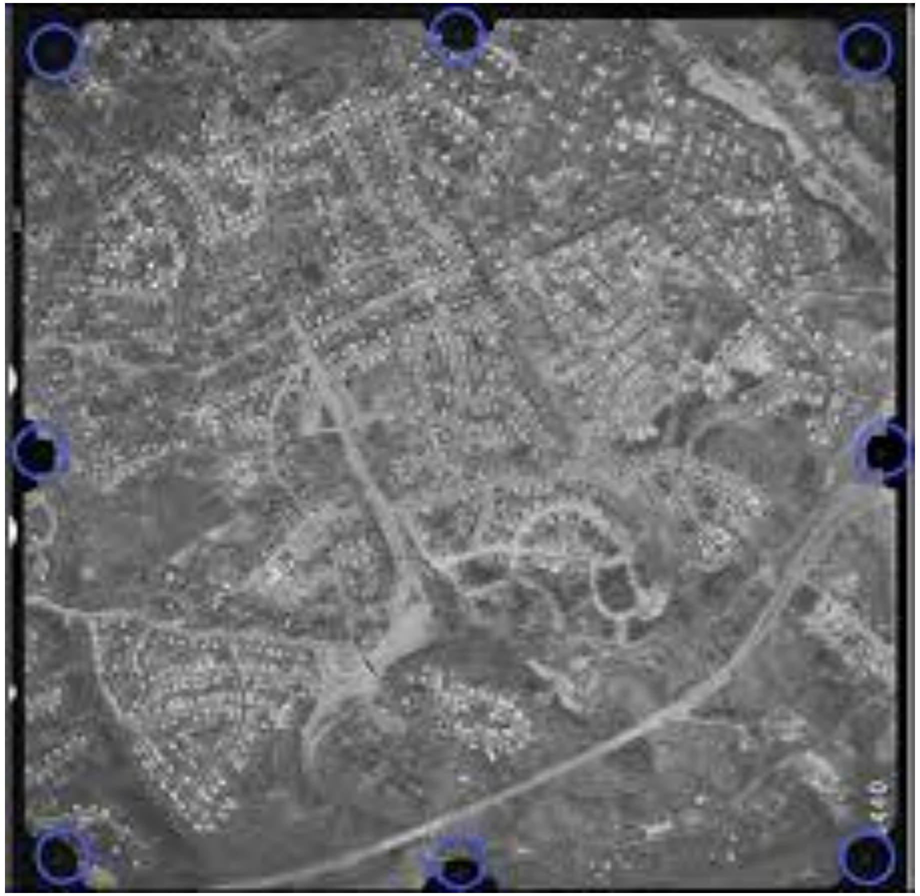
Focal Length

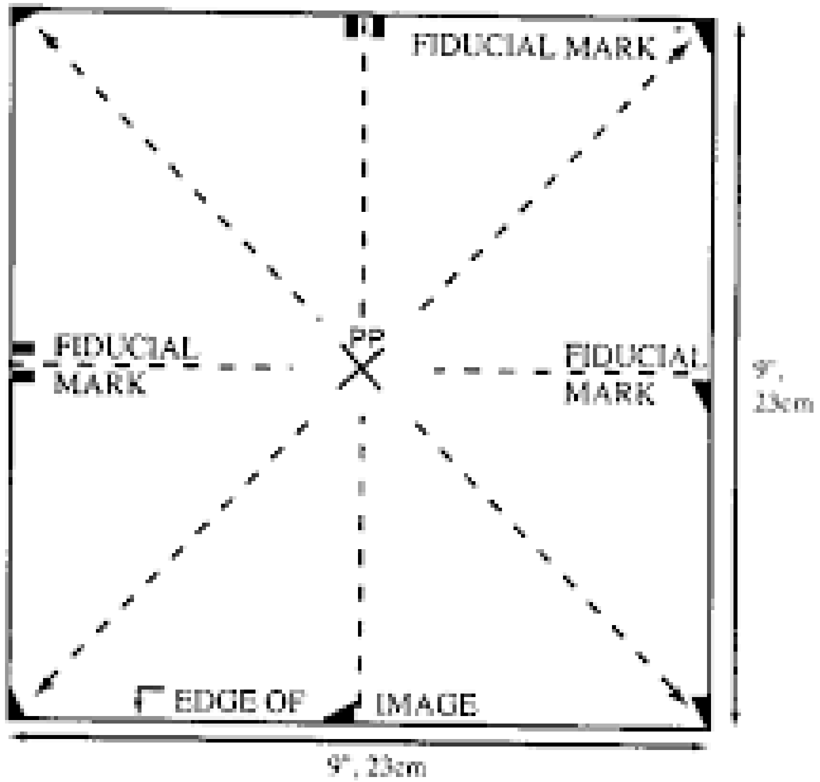
- The **focal length** is the distance between the camera lens and the image plane.
- It is one of the most important parameters in photogrammetry.
- Focal length influences the scale and coverage of aerial photographs.
- During calibration, the precise focal length of the camera is determined.



Fiducial Marks

- Fiducial marks are small reference marks placed on the edges of the photograph.
- They define the coordinate system of the image.
- Their positions relative to the camera lens are precisely known.
- These marks help establish the interior orientation during photogrammetric processing.




FIDUCIAL MARK
FIDUCIAL MARK
PP
FIDUCIAL MARK
EDGE OF IMAGE
9", 23cm
9", 23cm

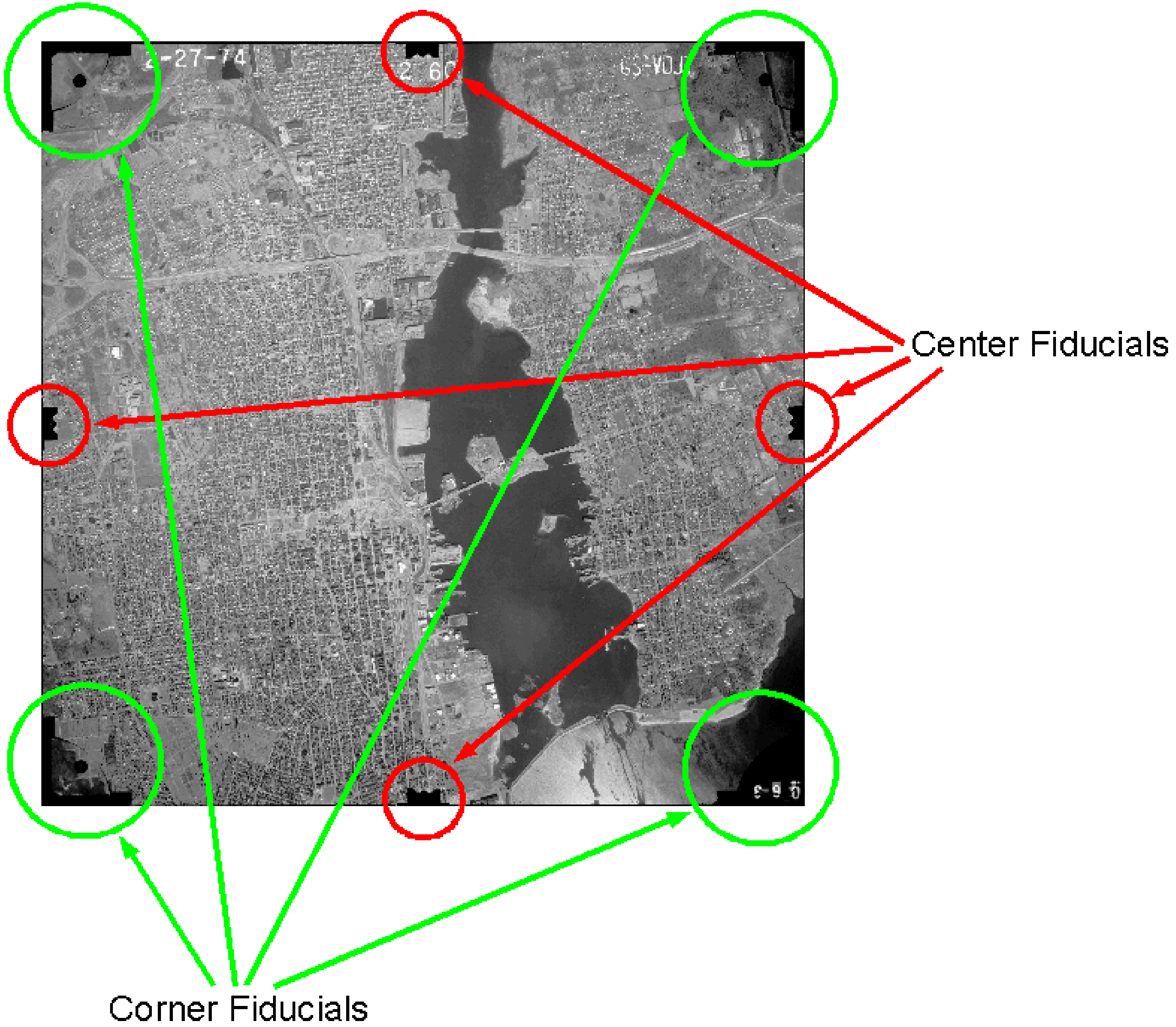

Center Fiducials
Corner Fiducials

Image Coordinate System

- The fiducial marks define the **image coordinate system**.
- This coordinate system is used to measure positions of points on the photograph.
- Image coordinates are later related to real-world coordinates.
- This relationship forms the basis of photogrammetric mapping.

Lens Distortion

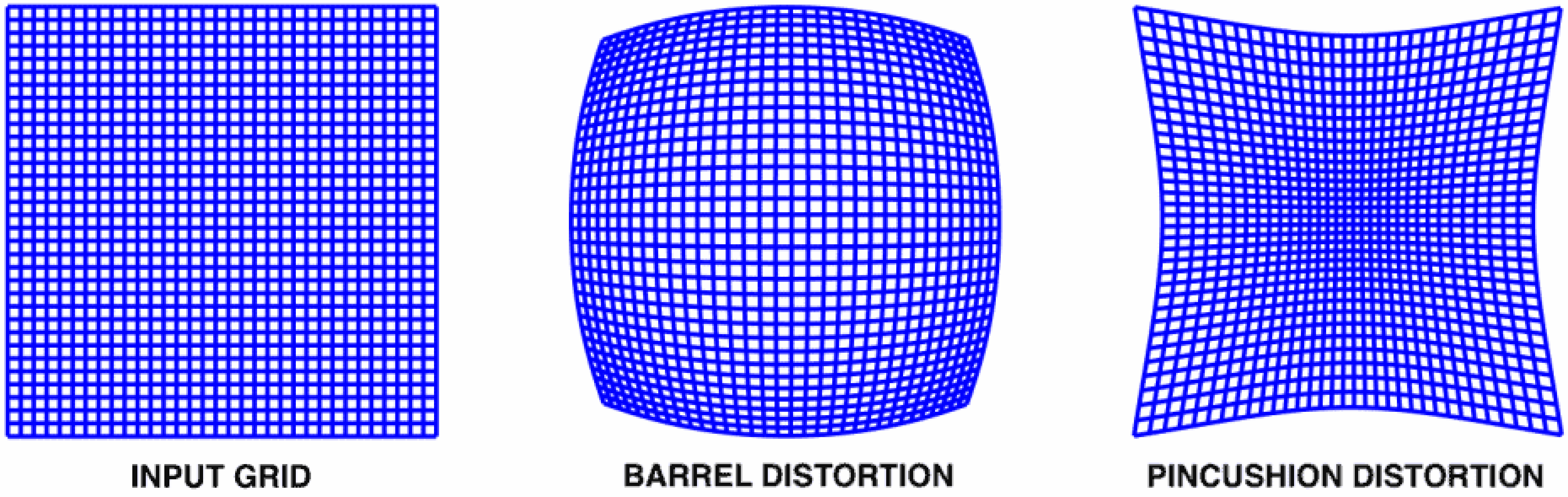
- No camera lens is perfectly accurate.
- Small distortions occur due to imperfections in the lens system.
- These distortions cause image points to shift slightly from their true positions.
- Calibration helps measure and correct these distortions.

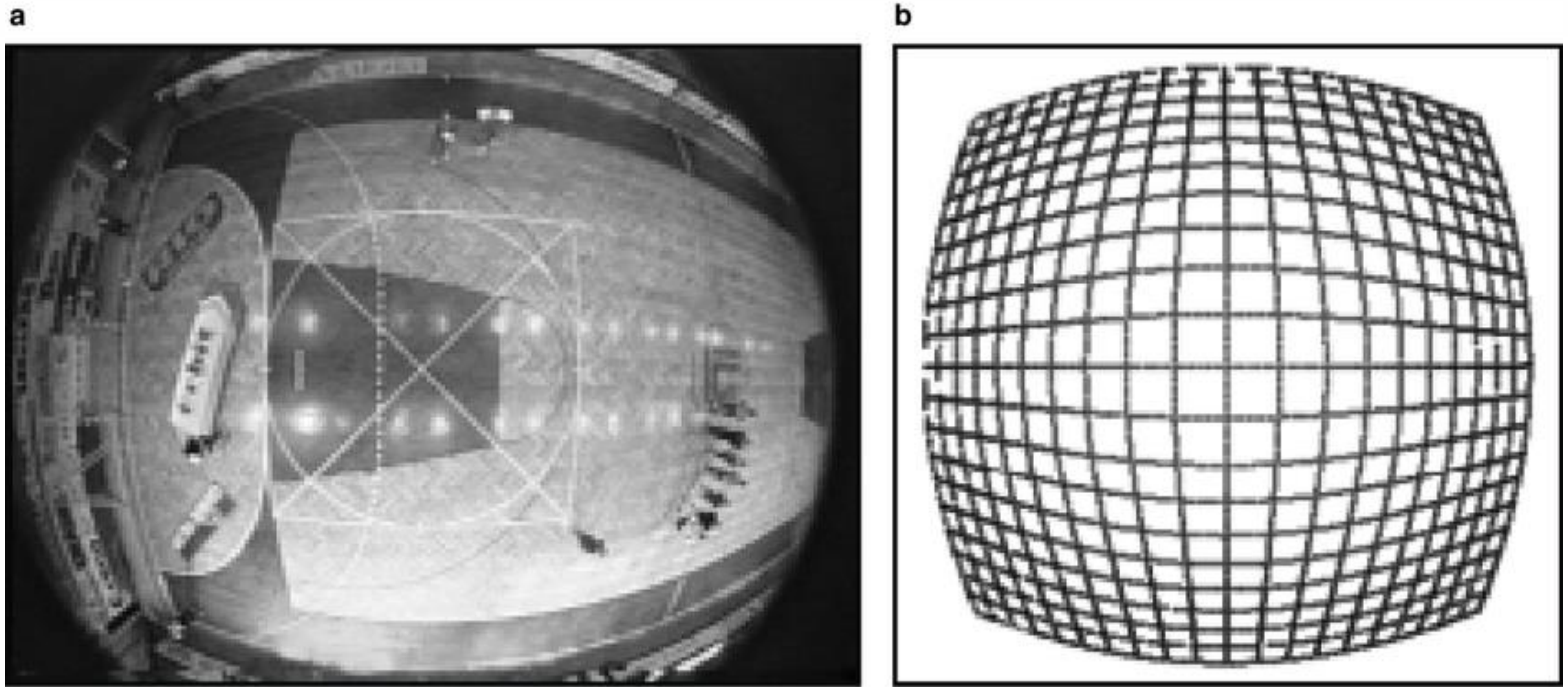
Types of Lens Distortion

- Two common types of lens distortion are:
- **Radial distortion**
- **Decentering distortion**
- These distortions are mathematically modeled during calibration.

Radial Distortion

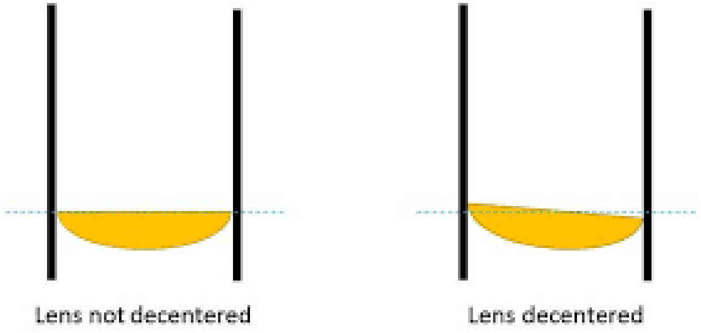
- Radial distortion occurs due to the shape of the lens elements.
- It causes image points to move outward or inward from the center of the image.
- This distortion increases toward the edges of the photograph.
- Calibration helps correct this effect.


INPUT GRID
BARREL DISTORTION
PINCUSHION DISTORTION


a
b

Decentering Distortion

- Decentering distortion occurs when lens elements are not perfectly aligned.
- It causes asymmetric distortion in the image.
- This distortion is usually smaller than radial distortion.
- However, it must still be corrected for accurate measurements.



Camera Calibration Report

- After calibration, a **calibration report** is produced.
- This report contains all important camera parameters.
- These parameters include focal length, principal point coordinates, and distortion coefficients.
- The report is used in photogrammetric processing.

Importance for Photogrammetry

- Accurate camera calibration improves measurement reliability.
- It allows precise determination of object positions and distances.
- Without calibration, photogrammetric results may contain systematic errors.
- Therefore, calibration is a fundamental step in photogrammetry.

Laboratory Calibration

- Most photogrammetric cameras are calibrated in specialized laboratories.
- The camera is tested under controlled conditions.
- Precise instruments are used to measure geometric parameters.
- Laboratory calibration ensures high accuracy.

Test Field Calibration

- Another method of calibration uses **test fields** on the ground.
- The test field contains many targets with known coordinates.
- Aerial photographs of this field are taken.
- The image measurements are compared with ground coordinates to determine calibration parameters.

In-Flight Calibration

- Sometimes calibration is performed during actual flight missions.
- Photographs are taken over known ground control points.
- Mathematical methods are used to estimate camera parameters.
- This method helps verify camera performance under real conditions.

Calibration Stability

- Camera calibration parameters should remain stable over time.
- However, factors such as temperature, vibration, and mechanical stress may change camera geometry.
- Therefore, cameras are periodically recalibrated.
- Regular calibration ensures accurate photogrammetric results.

Thank you!